

DEVAKUMAR A

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Summary

Computer Science Engineering graduate with full-stack development experience (Node.js, React.js, MongoDB) across three internships, and applied AI/ML project work spanning deep learning, NLP, and computer vision — including a Scopus-indexed research contribution. Seeking Software Engineer / Full Stack Developer roles to build and ship production-grade systems.

Internships

Web Framework Intern — Twilight IT Solution Pvt. Ltd.

Aug 2025 | Puducherry

- Built and styled responsive web pages using HTML, CSS, and JavaScript.
- Developed dynamic, interactive UI components using React.js, improving component reusability.
- Applied modern frontend workflows in a live, production-style environment.

Backend Development Intern — Askan Technologies

Jul 2025 | Puducherry

- Developed backend functionality using Node.js for server-side applications.
- Designed and managed databases with MongoDB and MongoDB Compass.
- Tested and debugged RESTful APIs using Postman, resolving issues prior to deployment.

Front-End Development Intern — Cognifyz Technologies

Feb 2025 – Mar 2025 | Remote

- Created responsive UIs using HTML, CSS, JavaScript, and Bootstrap.
- Collaborated directly with the design team through iterative feedback cycles to improve UX consistency.

Leadership

President — Mathoria Mathematics Club, RGCET

- Led planning and execution of academic events, competitions, and workshops for the club's student community.
- Mentored peer students in mathematical and analytical problem-solving.
- Built and shipped the club's management portal (see Projects) to digitize registration, events, and communication.

Projects

Deep Learning-Based EEG Communication Aid for Paralyzed Patients

- **Tech:** Python, TensorFlow, Keras, EEGNet, MNE [\[GitHub\]](#)
- Engineered a multi-task EEG-based Brain-Computer Interface combining intent detection, emotion recognition, and pain severity classification in a single deep learning architecture, achieving 89% overall accuracy.
- Built an EEG preprocessing pipeline (band-pass filtering, artifact removal, time-window segmentation) that improved signal quality and downstream classification performance.
- Designed an EEGNet-inspired model with spatial-temporal feature extraction and multi-head classification, reaching 86–92% accuracy across three independent healthcare tasks.

AquaCommandAI — AI-Powered Water Resource Intelligence Platform

- **Tech:** React.js, FastAPI, Python, Scikit-learn, PostgreSQL [\[GitHub\]](#)
- Built a full-stack platform integrating four ML models (crop recommendation, yield prediction, irrigation forecasting, reservoir risk) behind a FastAPI backend and React frontend.
- Designed RESTful APIs connecting model inference to real-time analytics dashboards for agricultural planning.
- Implemented data pipelines with Pandas and NumPy to feed predictive models from multi-source agricultural datasets.

Automated Resume Relevance Check System

- **Tech:** Python, Sentence Transformers, Hugging Face, Streamlit [GitHub]
- Built an NLP-based resume screening system using fine-tuned Sentence Transformer embeddings and cosine similarity to score candidate-job fit.
- Developed dual Student/HR dashboards in Streamlit with automated High/Medium/Low verdict generation.
- Integrated PDF and DOCX parsing pipelines to support real-world resume formats without manual preprocessing.

MovieVerse AI — Multi-Modal Genre Classification & Recommendation

- **Tech:** Python, TensorFlow, BERT, OpenCV, ResNet50 [GitHub]
- Built a multi-modal classification system combining transformer-based text embeddings (BERT) with CNN-based image features (ResNet50) to predict movie genres from plot, poster, and trailer data.
- Implemented a confidence-based multi-genre prediction engine, improving classification robustness over single-modality baselines.

Additional projects: QuantPulse — AI stock forecasting with LSTM/Regression (GitHub) | LinguaAssist — accessible speech-to-text/text-to-speech Android app (GitHub) | Mathoria Portal — club management web platform (GitHub)

Publications

Deep Learning-Based EEG Communication Aid for Paralyzed Patients

R. G. Suresh Kumar, **Devakumar A**, Roney Varughese Joseph, Sivasubramanian J
International Journal of Drug Delivery Technology (IJDDT), Volume 16, Issue 58s, 2026, pp. 235–241.
DOI: 10.25258/ijddt.16.58s.21
Scopus-Indexed Research Publication | Contributing Author

Technical Skills

Languages	Python, Java, C, C++, JavaScript, HTML, CSS
Full Stack	Node.js, React.js, MongoDB, FastAPI, Flask, Bootstrap
AI/ML	TensorFlow, Keras, Scikit-learn, Hugging Face, Sentence Transformers, NumPy, Pandas, NLP, Computer Vision
Databases	MySQL, PostgreSQL, SQLite
Tools & Cloud	Git, GitHub, Postman, Android Studio, Firebase, Oracle Cloud Infrastructure (OCI)
Soft Skills	Leadership, Team Management, Mentoring, Communication, Problem Solving

Education

2022 – 2026	B.Tech — Computer Science and Engineering, Rajiv Gandhi College of Engineering and Technology, Puducherry (CGPA: 7.63)
2022	HSC — Amalorpavam Higher Secondary School (82.8%)
2020	SSLC — Amalorpavam Higher Secondary School (78.2%)

Certifications

- Deep Learning — IIT Ropar (Gold + Elite, Top Rank)
- Python for Data Science — NPTEL (Silver + Elite)
- Oracle Cloud Infrastructure 2025 AI Foundations Associate — Oracle
- Oracle Cloud Infrastructure 2025 Generative AI Professional — Oracle
- Oracle Cloud Infrastructure 2025 Data Science Professional — Oracle
- Google Data Analytics — Coursera
- SQL (PostgreSQL) — IBM
- C++ Programming — IIT Bombay
- Career Essentials in Software Development — Microsoft
- Introduction to Data Science — Infosys Springboard
- Data Analytics — Novitech R&D Pvt. Ltd.
- Python Basics — Infosys Springboard
- Ethical Hacking: Cloud Computing Concepts — Infosys Springboard
- Drupal — IIT Bombay
- Intro to Front-End Development — Simplilearn
- Website UI/UX Designing Using ChatGPT — Simplilearn
- Excel for Beginners — Great Learning

Hackathons & Conferences

Smart India Hackathon 2025 | Smart India Hackathon 2024 | TATA Elxsi Innovation Challenge 2024 | GDG On Campus Solution Challenge 2025 (Hack2Skill) | TECHFEST 2025 | L'Oréal Sustainability Challenge 2025

Simulation

Software Engineering Job Simulation — JPMorgan Chase & Co. Jul 2024 – Aug 2024 | Virtual

- Completed a guided software development simulation involving modifications to an existing GitHub-hosted code-base.
- Interpreted inline comments and requirements to implement targeted code changes.
- Applied debugging, code optimization, and version control best practices.

Achievements

- NPTEL Deep Learning — IIT Ropar — **Elite + Gold** Certification (**Top 1% of Learners**, Score: 93%)
- NPTEL Python for Data Science — **Elite + Silver** Certification (Score: 81%)
- Trinity College London — Drumkit Grade 1 & Grade 2